

Higher-order faithfulness finds better Markov blankets, even on standard benchmarks



lklee.dev/uai26

High-Order Markov Blanket Discovery via a k -Order Relaxation of the Faithfulness Assumption



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Our Motivation

- **Markov blanket discovery** finds the minimal set shielding a target Y . Underpins Bayesian / Markov network structure learning.
- Nearly every method assumes **faithfulness**: an independence in the distribution P implies a separation in the graph G .
- But faithfulness can be **violated**, for example: on XOR/parity relations, and on finite samples

Our Contributions

- **k -order faithfulness**: assume an edge must still reveal itself in the distribution, but up to k **witness** variables may be needed to see it.
- **kOMB**: A proof of concept Grow-and-Shrink type algorithm. Provably recovers the blanket under k -order faithfulness.
- **Result**: kOMB $k = 1$ beats **all eight** baselines on **all four** benchmarks. On parity every baseline scores 0.03 where kOMB reaches **1.00**.

What is Faithfulness?

Every independence in the distribution P must appear as a separation in the graph G .

Example — **AND** gate between independent fair coins.

in the distribution P			in the graph G		
X	Z	Y	$X \perp\!\!\!\perp_P Z$	$\xRightarrow{\text{faithfulness}}$	$X \perp\!\!\!\perp_G Z$
0	0	0			
0	1	0			
1	0	0	$X \not\perp\!\!\!\perp_P Y$	$\xleftarrow{\text{contra-positive}}$	$X \not\perp\!\!\!\perp_G Y$
1	1	1	$Z \not\perp\!\!\!\perp_P Y$		$Z \not\perp\!\!\!\perp_G Y$

Generally under the faithfulness assumption:

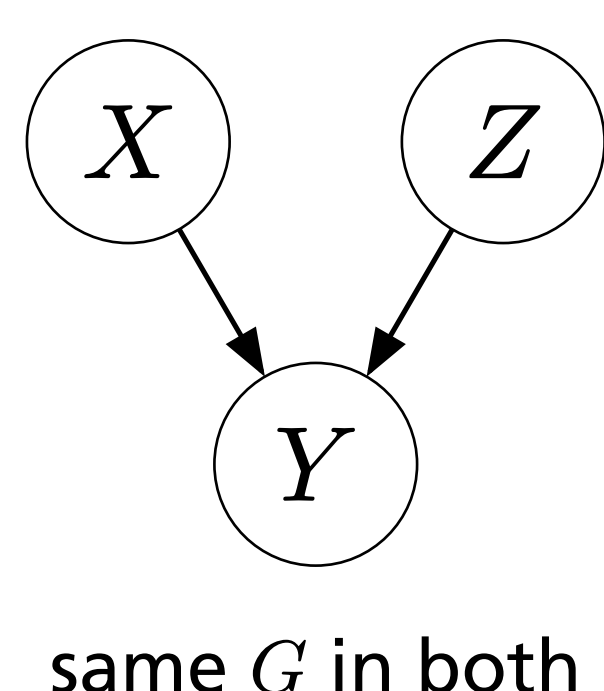
$$X \perp\!\!\!\perp_P Y \mid S \implies X \perp\!\!\!\perp_G Y \mid S \quad \text{for all disjoint } X, Y, S$$

$$X \not\perp\!\!\!\perp_P Y \mid S \iff X \not\perp\!\!\!\perp_G Y \mid S \quad \text{for all disjoint } X, Y, S$$

When Faithfulness is Violated

There are two ways the faithfulness assumption can be violated:

- **Empirical** — faithful P and G , but unlucky sample; more data fixes it.
- **Structural** — P itself is unfaithful to G ; no sample size helps.



Empirical

Noisy sample from **AND**

X	Z	Y	
0	0	0	$Y \perp\!\!\!\perp_P X$
0	1	1	$\Downarrow$
1	0	0	$Y \not\perp\!\!\!\perp_G X$
1	1	1	

One noisy flip makes Y appear independent of X

Structural

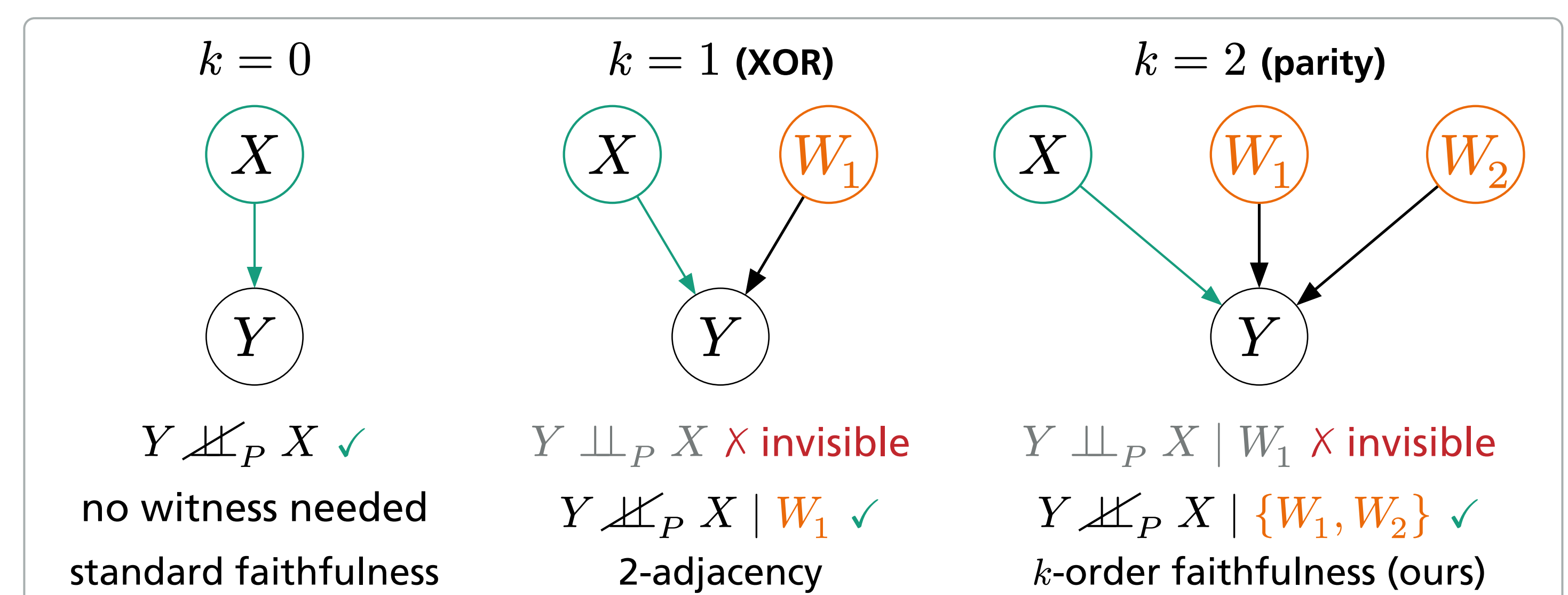
Full samples from **XOR**

X	Z	Y	
0	0	0	$Y \perp\!\!\!\perp_P X$
0	1	1	$\Downarrow$
1	0	1	$Y \not\perp\!\!\!\perp_G X$
1	1	0	

Visible only as $Y \not\perp\!\!\!\perp_P X \mid Z$ — needs order $k = 1$.

k -Order Faithfulness

An edge might need up to k variables to be visible in P .

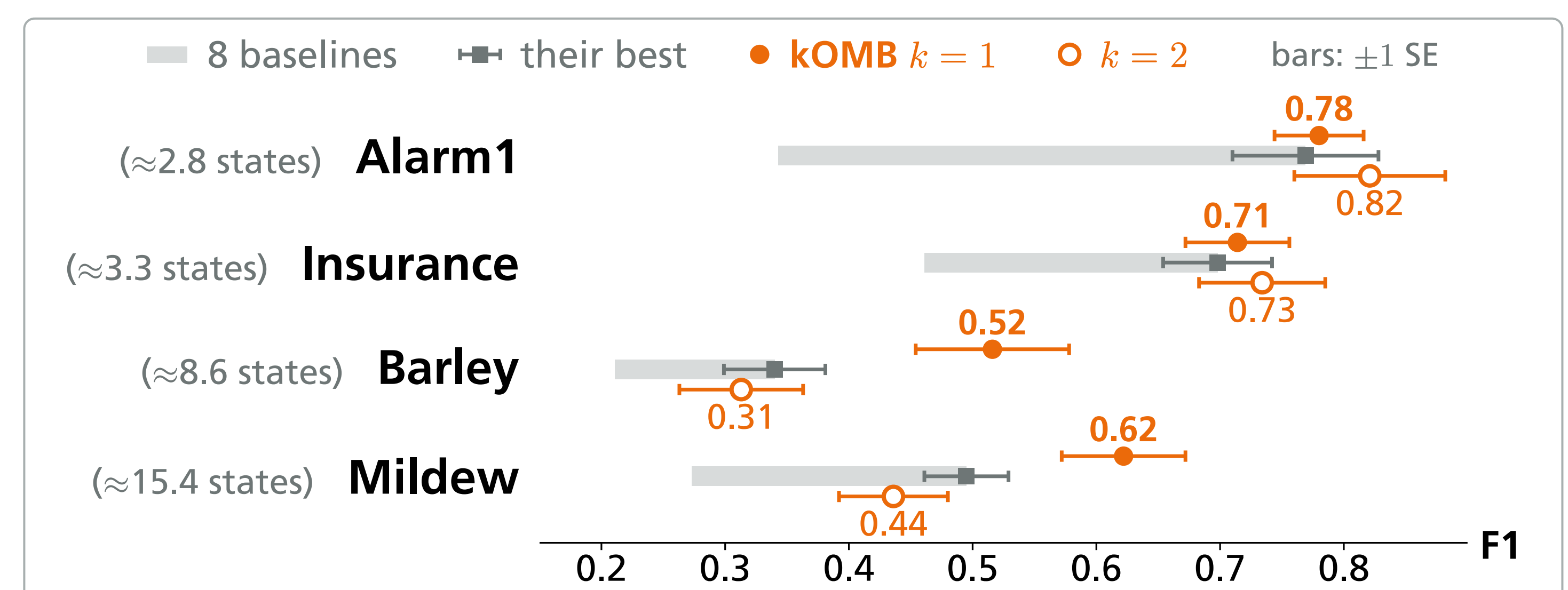


Increasing order admits edges invisible on lower orders.
 Generally, parity over $k + 2$ variables needs order k .

Results

- **Recovers Parity Perfectly.** On synthetic parity datasets, every baseline scores 0.03 while kOMB with $k = 2$ reaches **1.00**.
- **kOMB $k = 1$ beats every baseline on all benchmarks.**

Mean F1 on benchmark networks, $N = 5000$.



- **A bigger search budget cuts both ways:** it wins on Alarm1 and Insurance, but on high-cardinality nets the independence tests lose power, and $k = 1$ wins.
- **The cost.** Runtime grows with the budget: $k = 1$ takes 5–58 s, $k = 2$ takes 14–336 s, $k = 3$ hits a 30-min cap.